

CHALLENGES & SOLUTIONS REPORT

Introduction

During the internship, our team explored various Artificial Intelligence and Computer Vision technologies related to fashion recommendation systems and virtual try-on applications. While researching and analyzing different models, workflows, and datasets, we encountered several technical and project-related challenges. Understanding these challenges and identifying potential solutions helped us gain deeper insights into the practical implementation of AI-powered fashion systems.

1. Human Pose Variations

Challenge:

One of the major challenges in virtual try-on systems is handling different body poses. Users may stand, sit, bend, or take pictures from different angles, making it difficult for AI models to accurately detect body landmarks and measurements. Variations in lighting conditions and image quality can further affect pose detection accuracy.

Solution:

To address this challenge, we studied pose estimation models such as MediaPipe and OpenPose. These models provide body landmark detection and skeleton tracking capabilities that help identify key body points. By improving landmark detection and using robust pose estimation techniques, virtual try-on systems can achieve better alignment and fitting accuracy.

2. Clothing Alignment and Garment Fitting

Challenge:

Properly aligning a garment with a user's body is one of the most important aspects of virtual try-on systems. Differences in body shapes, clothing sizes, and user poses can make garment placement difficult and may result in unrealistic outputs.

Solution:

We explored garment mapping and warping techniques used in modern virtual try-on architectures. Research on HR-VITON and related virtual try-on systems demonstrated how AI can transform and align garments according to body shape and pose information. These techniques significantly improve fitting realism and garment placement accuracy.

3. Segmentation Accuracy

Challenge:

Virtual try-on systems require accurate separation of clothing and body regions from the background. Inaccurate segmentation may lead to distorted outputs, overlapping garments, or incomplete clothing extraction.

Solution:

We researched segmentation models such as U2Net and Detectron2. These models are capable of identifying clothing regions and generating detailed masks for garment extraction. Improved segmentation workflows help create cleaner inputs for virtual try-on systems and enhance overall output quality.

4. Dataset Availability and Quality

Challenge:

AI models require large and diverse datasets for training and evaluation. Finding datasets containing multiple clothing categories, body types, poses, and annotations can be challenging. Limited or imbalanced datasets may affect model performance and generalization.

Solution:

To overcome this challenge, we explored multiple datasets including DeepFashion, DeepFashion2, and VITON-HD. Studying various datasets allowed us to understand the importance of data diversity and proper dataset organization for achieving reliable AI performance.

5. Real-Time Processing Constraints

Challenge:

Virtual try-on systems involve several computationally intensive processes such as pose detection, segmentation, garment alignment, and recommendation generation. Performing these operations in real time can be difficult, especially on devices with limited resources.

Solution:

We analyzed lightweight AI models and optimized workflows used in modern fashion technology applications. Efficient model selection, preprocessing techniques, and streamlined AI pipelines can help reduce processing time while maintaining acceptable performance.

6. Integration of Multiple AI Modules

Challenge:

Building a complete virtual try-on platform requires integrating several AI modules, including body detection, pose estimation, segmentation, garment mapping, recommendation systems, and user interfaces. Coordinating these modules can be complex due to dependencies and data flow requirements.

Solution:

To address this challenge, we designed a structured AI workflow that connects all major components into a unified pipeline. Defining clear input-output relationships between modules helps improve system organization and future scalability.

7. Research and Technology Selection**Challenge:**

Numerous AI models, frameworks, and technologies are available for fashion applications. Selecting appropriate solutions for specific tasks requires extensive research and comparison.

Solution:

Throughout the internship, we conducted comparative studies of various models and technologies. Evaluating their advantages, limitations, and use cases helped us understand which approaches are most suitable for different fashion AI applications.

Conclusion

The challenges encountered during this internship provided valuable learning opportunities and practical insights into the development of AI-powered fashion systems. By studying existing solutions and analyzing modern AI workflows, we gained a better understanding of how virtual try-on and recommendation systems can be improved to deliver more accurate, realistic, and user-friendly experiences. These findings can contribute to the future development of intelligent fashion platforms such as Raritone.